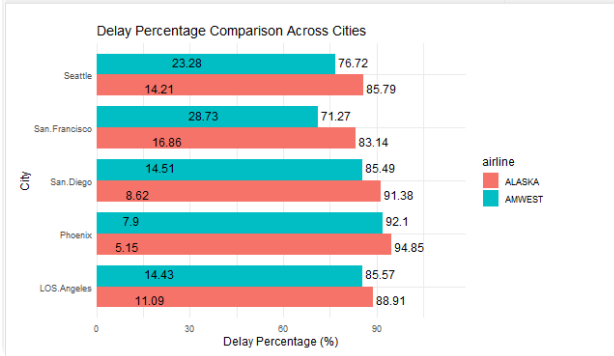
```r
ggplot(city_compare, aes(x = percentage, y = City, fill = airline)) +
  geom_col(position = "dodge", width = 0.8) +
  geom_text(aes(label = percentage),
            position = position_dodge(width = 1),
            hjust = -0.1,
            size = 4 )+
  labs(title = "Delay Percentage Comparison Across Cities",
       x = "Delay Percentage (%)",
       y = "City") +
  theme_minimal() +
  scale_x_continuous(expand = expansion(mult = c(0, 0.25)))
```



how to fix

Thought for 41s ›

You're plotting **both delay % and non-delay %** for each `City + airline`, so `geom_col()` is drawing **two rows per group**. That is why you see numbers like:

- `14.21` and `85.79`
- `23.28` and `76.72`

Those pairs add to **100%**.

Fix

For a **Delay Percentage Comparison** chart, keep **only the delay rows** before plotting.

Check the problem

R

If you get `n = 2` for each group, that means each city-airline pair has 2 rows in the data.

Correct version

Replace `status` and `"delayed"` with your real column/value names.

```
city_delay <- city_compare %>%
  filter(status == "delayed")

ggplot(city_delay, aes(x = percentage, y = City, fill = airline)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.8) +
  geom_text(
    aes(label = round(percentage, 2)),
    position = position_dodge(width = 0.8),
    hjust = -0.1,
    size = 4
  ) +
  labs(
    title = "Delay Percentage Comparison Across Cities",
    x = "Delay Percentage (%)",
    y = "City"
  ) +
  theme_minimal() +
  scale_x_continuous(
```

R

Why this works

Your chart should have **one row per** `City + airline` for delay rate only.

So it should look like:

City	airline	percentage
Seattle	ALASKA	14.21
Seattle	AMWEST	23.28

Not this:

City	airline	status	percentage
Seattle	ALASKA	delayed	14.21

City	airline	status	percentage
Seattle	ALASKA	on_time	85.79

If you want to verify quickly

Run this:

```
city_compare %>%
```

R

If you paste your `city_compare` code, I can point out the exact line causing it.

what if i wanna show both ontime and delayed

Thought for a few seconds

Then don't label it as only **Delay Percentage**. Right now your chart is actually showing **both statuses**.

Option 1 — Best choice: stacked 100% bar

If you want to show **ontime + delayed together**, use a stacked bar so each airline in each city adds to 100%.

```
ggplot(city_compare, aes(x = City, y = percentage, fill = status)) +
  geom_col() +
  facet_wrap(~ airline) +
  geom_text(aes(label = round(percentage, 2)),
            position = position_stack(vjust = 0.5),
            size = 4) +
  labs(
```

```
    title = "On-time vs Delayed Percentage Across Cities",
    x = "City",
    y = "Percentage (%)",
    fill = "Status"
```

R

What this does

- each bar = one city
- bar split into **ontime** and **delayed**
- separate panel for each airline

This is usually the cleanest way if both percentages sum to 100.

Option 2 — Keep horizontal bars and show both separately

If you want **4 bars per city** (`ALASKA delayed` , `ALASKA ontime` , `AMWEST delayed` , `AMWEST ontime`), map `status` too.

```
ggplot(city_compare, aes(x = percentage, y = City, fill = interaction(airline, st
  geom_col(position = position_dodge(width = 0.9), width = 0.8) +
  geom_text(
    aes(label = round(percentage, 2)),
    position = position_dodge(width = 0.9),
    hjust = -0.1,
    size = 3.5
  ) +
  labs(
    title = "On-time and Delayed Percentages Across Cities",
    x = "Percentage (%)",
    y = "City",
    fill = "Airline / Status"
  ) +
```

R

Problem with this version

It works, but it gets crowded fast.

Option 3 — Facet by status

This is often the easiest to read.

```
ggplot(city_compare, aes(x = percentage, y = City, fill = airline)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.8) +
  geom_text(
    aes(label = round(percentage, 2)),
    position = position_dodge(width = 0.8),
    hjust = -0.1,
    size = 3.5
  ) +
  facet_wrap(~ status) +
  labs(
    title = "On-time and Delayed Percentages Across Cities",
    x = "Percentage (%)",
    y = "City"
  ) +
```

R

This gives

- one panel for **ontime**
- one panel for **delayed**
- within each panel, compare airlines by city

Recommendation

For this assignment, I'd use:

- **facet by status** if you want easy comparison
- **stacked bar** if you want to emphasize that ontime + delayed = 100

Also change your title from:

- `Delay Percentage Comparison Across Cities`

to something like:

- `On-time and Delayed Percentage Across Cities`

If you want, paste your actual column names (`status` , `delay` , `type` , etc.), and I'll rewrite the exact final code for your dataset.

Week5a.pdf
PDF

Have I completed my assignment as required?

Thought for 30s

Based on **the PDF you submitted**, **yes — you appear to have completed the core assignment requirements**. I cannot verify against the full original prompt unless

ChatGPT 5.2 Thinking

What you clearly completed

- **Data import + cleaning + reshape from wide to long**: you read the CSV, renamed fields, filled the airline values, and used `pivot_longer()` to produce a tidy long table. Week5a
- **Overall airline comparison**: you computed overall on-time and delayed percentages for both airlines and graphed them. Week5a Week5a
- **City-level comparison**: you calculated city-by-city percentages for each airline and plotted them. Week5a Week5a Week5a
- **Simpson's paradox conclusion**: you explicitly stated that ALASKA looks better within each city, while AMWEST looks better overall, and identified that as Simpson's paradox. Week5a

My judgment

Academically: probably yes.
Your submission demonstrates the required transformation, comparison, and interpretation. The logic is coherent, and the conclusion matches the data shown in the PDF. Week5a Week5a

Small issues to fix before final submission

These are **polish issues**, not major analytical problems:

1. **Typo:** `MWEST` should be `AMWEST`. Week5a
2. **Typo:** `Compare by each cites` should be `Compare by each city` or `Compare by city`. Week5a
3. Your city chart shows **both on-time and delayed**, but the paragraph below describes it as a **delay-rate comparison**. That is slightly mismatched. Either:
   - keep the current stacked chart and describe it as **"on-time vs delayed by city"**, or
   - plot only delayed percentages if you want the paragraph to stay focused on delay rate. Week5a Week5a
4. If the instructor required the **data file itself** in addition to the PDF/Rmd, make sure you also submit that separately. Your PDF references the source CSV, but I cannot confirm whether the submission portal also expects the raw file upload. Week5a

Bottom line

Yes, the assignment looks complete enough to submit.
I would just fix the wording/typos above so it looks cleaner and more professional.

PDF: Week5a.pdf

Paste the assignment prompt/rubric if you want a stricter **"meets every requirement or not"** check.

Sources